



Name: \_\_\_\_\_ Cohort/Batch: \_\_\_\_\_ Date: \_\_\_\_\_ Score: \_\_\_\_\_

## CODEIGNITER PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 Q&A on MVC, routing, Query Builder, validation, and testing

TOTAL: 10 QUESTIONS

**Q1** What is CodeIgniter and how does it implement MVC?

Threat Model

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**Q6** How does CodeIgniter 4 handle sessions differently from PHP native sessions?

Observability

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**Q2** How does CodeIgniter 4 routing work?

Architecture

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**Q7** How do you handle file uploads in CodeIgniter 4?

Lifecycle

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**Q3** How does the CodeIgniter Query Builder prevent SQL injection?

Configuration

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**Q8** What is a CodeIgniter RESTful Resource Controller?

Compliance

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**Q4** What are CodeIgniter 4 Models and Entity classes?

Hardening

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**Q9** How does CodeIgniter 4 Filters work?

Testing

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**Q5** How does CodeIgniter validation work?

SSO / IdP

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**Q10** How do you test a CodeIgniter 4 controller?

Tuning

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## ANSWER KEY &amp; EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls &amp; reference

VERIFIED SOLUTIONS

## A1 CodeIgniter 4 is a lightweight PHP

Mitigation

CodeIgniter 4 is a lightweight PHP framework. Controllers in `app/Controllers` extend `BaseController` and handle HTTP logic. Models in `app/Models` extend `Model` and interact with the database. Views in `app/Views` are PHP files rendered with `return view('name', data)`.

## A6 CI4's session library wraps native sessions

Observability

CI4's session library wraps native sessions but adds configurable drivers (files, database, Redis, Memcached). Sessions are signed with an encryption key to prevent tampering. `set_flashdata('key', val)` stores one-request messages. Regeneration is automated.

A2 Routes are defined in `app/Config/Routes.php`:

Primitives

Routes are defined in `app/Config/Routes.php`: `$routes->get('users/{:num}', 'UserController::show/$1')`. `$routes->resource('users')` creates CRUD routes automatically. Auto-routing (deprecated) mapped URLs to controllers by convention.

A7 Access via `$file =`

Automation

Access via `$file = $this->request->getFile('avatar')`. Call `$file->isValid()` and `$file->hasMoved()` before moving. `$file->move(WRITEPATH.'uploads', $file->getRandomName())` stores it. Validate MIME type with `getClientMimeType()` and check file size against `maxSize`.

## A3 Query Builder methods accept raw values

Hardening

Query Builder methods accept raw values that are automatically escaped and parameterized. `$db->where('id', $id)->get('users')` generates a prepared statement. Never concatenate user input into query strings; use Query Builder or `$db->query("SELECT * FROM u WHERE id=?", [$id])`.

A8 `php artisan make:controller UserController`

Governance

`php artisan make:controller UserController --rest` generates index, show, create, store, edit, update, delete stubs. `$routes->resource('users', ['controller'=>'UserController'])` maps HTTP verbs to these methods. Override only the methods your resource needs.

## A4 A Model extending CI4's Model gets

Gotchas

A Model extending CI4's Model gets CRUD helpers (`find`, `insert`, `update`, `delete`) and built-in validation. An Entity class maps a database row to a typed PHP object with casting (`int`, `bool`, `datetime`). Set protected `$returnType = UserEntity::class` in the model.

## A9 Filters (like middleware) run before or

Assessment

Filters (like middleware) run before or after controllers. Define a filter class with `before()` and `after()` methods. Register in `app/Config/Filters.php` under `$aliases`. Apply globally or per-route: `$routes->get('admin/*', 'AdminController', ['filter' => 'auth'])`.

A5 Use `$this->validate($rules)` in a controller, where

Federation

Use `$this->validate($rules)` in a controller, where `$rules` is an array like `['email' => 'required|valid_email']`. Define reusable rule sets in `app/Config/Validation.php`. `$this->validator->getErrors()` returns field-level error messages after a failed validation.

A10 Extend `CIUnitTestCase`. Use `$result =`

Optimization

Extend `CIUnitTestCase`. Use `$result = $this->withUri('http://example.com/users/1')->controller(UserController::class)->execute('show/1')`. `$result->assertStatus(200)` and `$result->assertSee('Alice')` check the response without starting an HTTP server.